ITM UNIVERSITY GWALIOR

School of Computing and Intelligent Systems

PROJECT-BASED LEARNING (PBL)

MINI PROJECT REPORT

Analysis of Student Attendance vs Academic Performance

Project detail	Information
Course	Data Analytics Using Python (CSL-0312)
Program	B.Tech. CSE (Artificial Intelligence & Machine Learning)
Semester / Section	III / F
Academic Session	July–December 2026
Domain	Student Performance & Learning Analytics
Team Leader	P. Chethan (BETN1A125194)
Team Members	G. Nandini (BETN1A125164); J. Durgeshwar (BETN1A125167)
Faculty Mentor	Dr. Sanjay Jain
Submission date	15/10/2026

Dataset used: students_500_dataset_updated.csv

1. Introduction & Problem Statement

Student attendance is routinely recorded by educational institutions, while academic performance is assessed through internal assessments, assignments, and final examinations. These records can be reviewed separately, which may make it difficult to understand whether attendance levels are associated with differences in academic outcomes. This project studies the relationship between student attendance and final examination marks using a structured dataset and Python-based data analytics.

The main problem addressed is whether students with different attendance levels show different patterns in final marks, and how strong the overall relationship between attendance and final marks is in the selected data. The analysis also considers internal marks, study hours, assignment scores, and previous-semester scores as contextual variables. The project uses descriptive statistics, group-wise comparisons, correlation analysis, and visualizations to summarize the data and communicate patterns clearly.

The scope is limited to exploratory and basic statistical analysis. Results describe associations within these records and do not establish that attendance alone causes higher or lower marks. Other influences may also affect performance.

Project objectives

• Examine attendance and academic-performance distributions.

• Clean and prepare the dataset for analysis.

• Measure the relationship between attendance and final marks.

• Compare final marks across Low, Medium, and High attendance groups.

• Communicate patterns and limitations using charts and statistical summaries.

2. Dataset Description & Source

Item	Description
Dataset name	Student Performance Dataset
File used	students_500_dataset_updated.csv
Source stated in synopsis	Kaggle https://docs.google.com/spreadsheets/d/1YYSdovK5C4CFHK7zK44vV3E8W_xpTvWX/edit?usp=sharing&ouid=115098169434384855124&rtpof=true&sd=true
Type	Public, as identified in approved synopsis
Records / columns	500 / 9
Quality checks	0 missing values; 0 duplicate rows
Primary variables	Attendance and FinalMarks
Other attributes	index, Name, InternalMarks, StudyHours, AssignmentScore, PreviousSemesterScore, Attendance_Gro

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Attribute overview

Attribute	Role
index	Row identifier; not used as a performance measure
Name	Student label; excluded from numeric statistics
Attendance	Attendance value/percentage
InternalMarks	Internal assessment score
FinalMarks	Primary outcome variable
StudyHours	Study hours recorded
AssignmentScore	Assignment score
PreviousSemesterScore	Previous-semester score
Attendance_Group	Low, Medium, or High group

3. Data Cleaning & Preprocessing

The CSV was loaded with Pandas and checked for dimensions, columns, missing values, and duplicate rows. It contained 500 rows and 9 columns, with no missing values and no fully duplicated rows. Numeric fields were reviewed using descriptive statistics. Name and index were retained as identifiers but excluded from correlation calculations. No imputation or duplicate removal was needed.

Attendance_Group was used for group-wise comparisons. The team should verify the attendance thresholds used to assign groups. Normalization, scaling, and encoding were not required for the descriptive and correlation analyses presented.

Representative code

import pandas as pd
from scipy import stats

df = pd.read_csv("students_500_dataset_updated.csv")
print(df.shape)
print(df.isna().sum())
print("Duplicate rows:", df.duplicated().sum())
r, p = stats.pearsonr(df["Attendance"], df["FinalMarks"])

4. Exploratory Data Analysis (EDA)

Attendance mean = 77.90 (SD 11.27); FinalMarks mean = 71.46 (SD 12.53; median 71.55).

Variable	Mean	Median	SD	Min	Max
Attendance	77.90	78.20	11.27	45.00	100.00
InternalMarks	68.40	68.40	14.45	30.00	100.00
FinalMarks	71.46	71.55	12.53	37.60	100.00
StudyHours	6.84	6.80	1.34	2.70	10.00
AssignmentScore	41.60	41.20	11.41	20.00	74.90
PreviousSemesterScore	53.23	53.35	13.84	25.00	89.20

Group	n	Mean attendance	Mean final marks	SD final marks
Low	130	63.70	66.38	12.40
Medium	233	77.69	71.06	11.76
High	137	91.74	76.96	11.81

5. Data Visualization & Observations

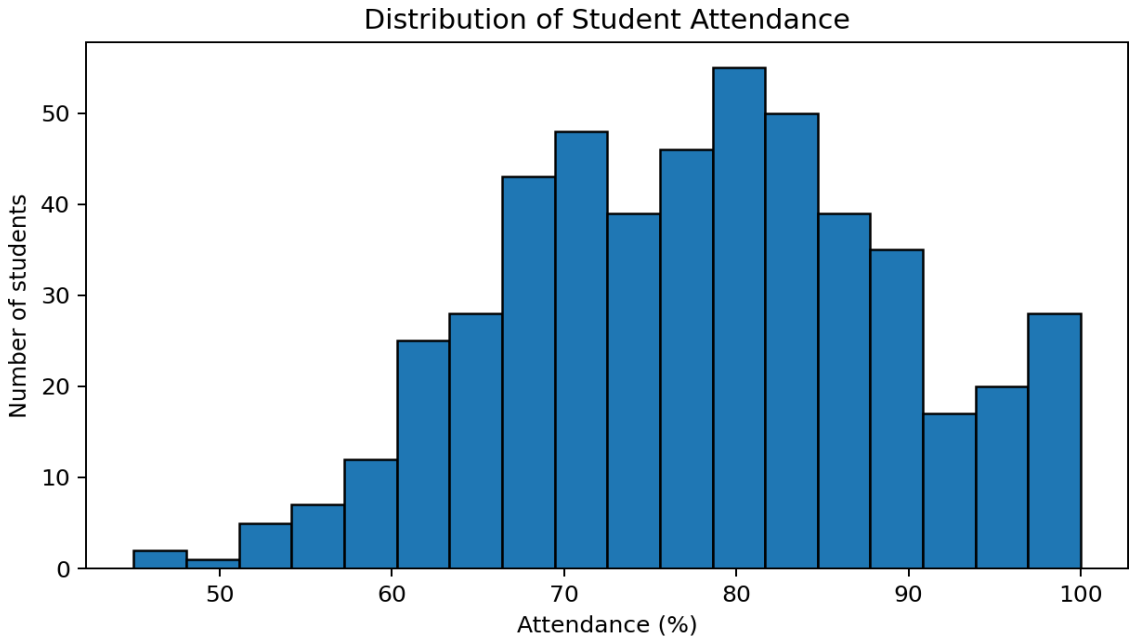


Figure 1. Attendance distribution. The histogram displays the spread and concentration of recorded attendance values.

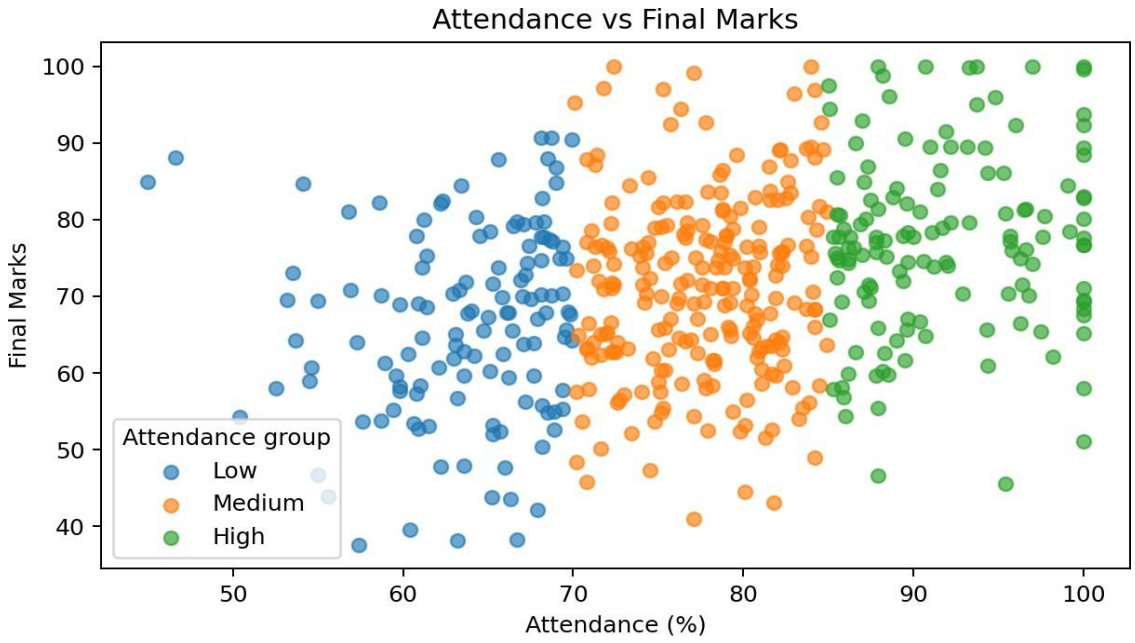


Figure 2. Attendance vs FinalMarks. The upward tendency is consistent with a positive Pearson correlation (r = 0.324), although individual variation remains.

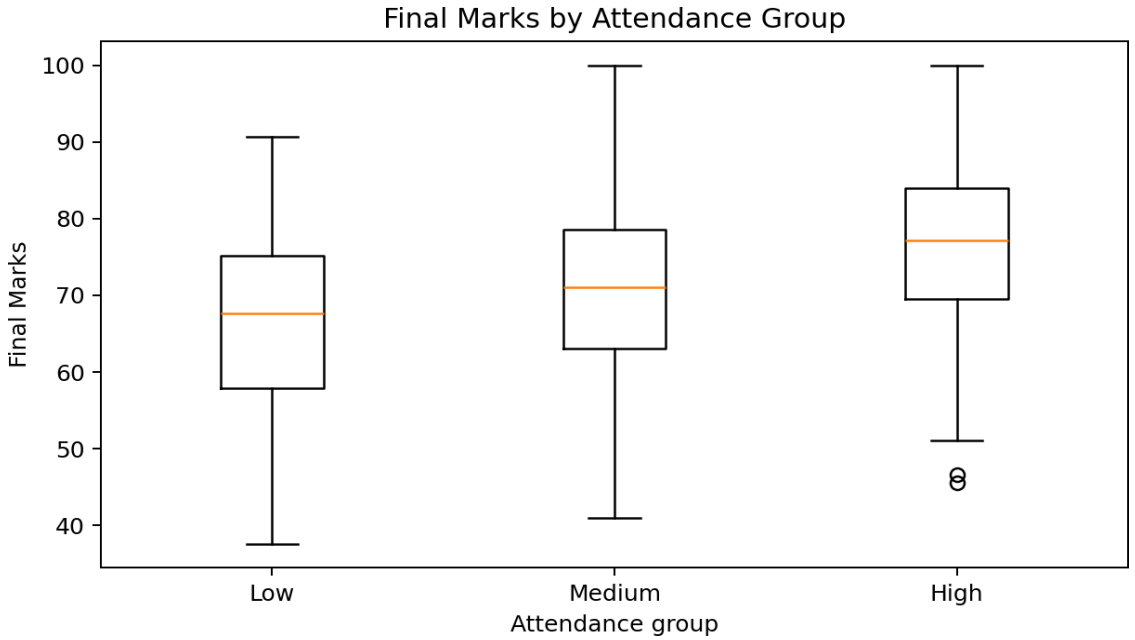


Figure 3. Final marks by attendance group. The High group has a higher average and median than the Medium and Low groups, with overlap between groups.

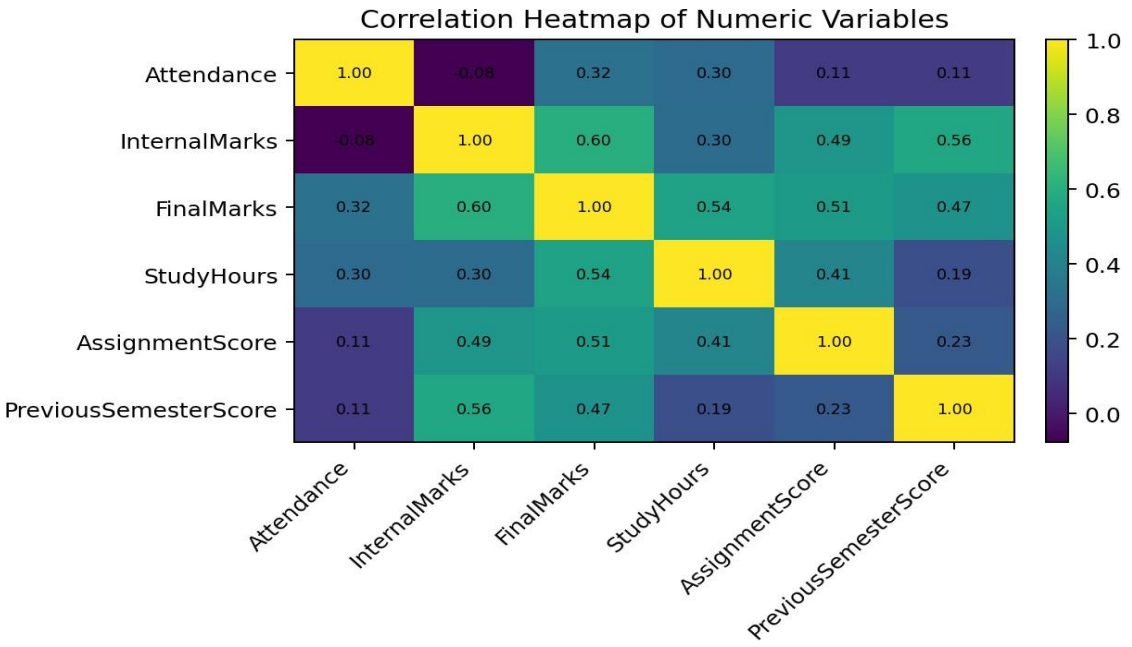


Figure 4. Correlation heatmap. InternalMarks and StudyHours are positively associated with FinalMarks. Correlation does not prove causation.

Observations on Figures 1–4

- **Figure 1 (Attendance distribution):** The histogram is roughly bell-shaped and peaks at about 80%. Most students fall between roughly 60% and 90% attendance, only a few attend below 55%, and a small rise appears at the 100% end of the scale.
- **Figure 2 (Attendance vs FinalMarks):** The points drift upward from left to right, which matches the positive correlation (r = 0.324). The spread around this trend is wide, so students with similar attendance can score very differently. Low-attendance students (blue) sit on the left and High-attendance students (green) on the right.
- **Figure 3 (Final marks by attendance group):** The median rises from 67.70 (Low) to 71.10 (Medium) to 77.20 (High), and the whole box shifts upward in the same order. The boxes overlap, so high attendance does not guarantee high marks and low attendance does not guarantee low marks. Two low outliers (about 46) appear in the High group.
- **Figure 4 (Correlation heatmap):** InternalMarks (0.60), StudyHours (0.54), AssignmentScore (0.51) and PreviousSemesterScore (0.47) have stronger links with FinalMarks than Attendance (0.32). Attendance is positively linked with StudyHours (0.30), but shows almost no link with InternalMarks (-0.08) and only a very weak link with AssignmentScore and PreviousSemesterScore (0.11 each).

Additional visualizations and observations

The charts below extend the analysis shown in Figures 1–4. Each figure is followed by a short observation.

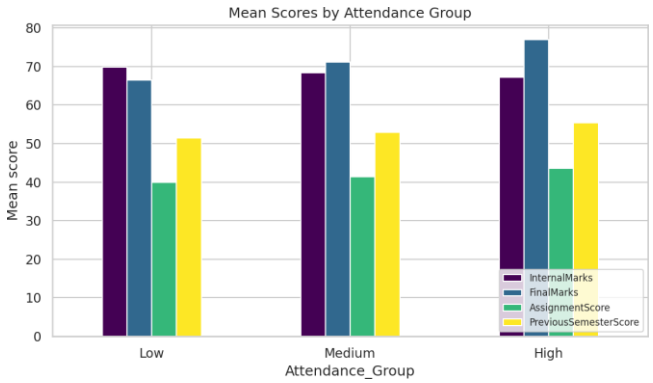


Figure 5. Mean scores by attendance group. The grouped bars compare the average InternalMarks, FinalMarks, AssignmentScore and PreviousSemesterScore of each group.

Observation: Mean FinalMarks rise from 66.38 (Low) to 71.06 (Medium) to 76.96 (High). AssignmentScore and PreviousSemesterScore also rise from the Low to the High group, while InternalMarks stays almost flat (about 69.8, 68.3 and 67.2), so the group differences appear mainly in the final examination.

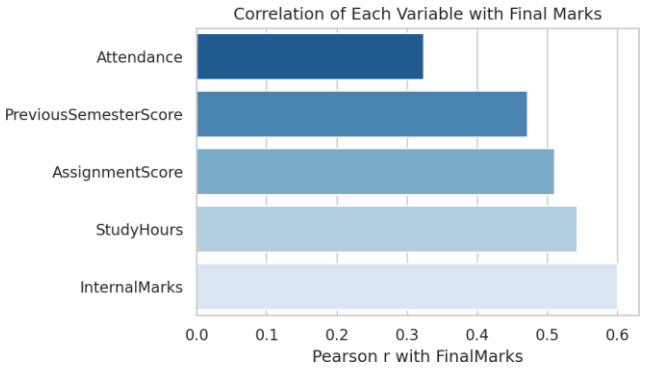


Figure 6. Correlation of each variable with FinalMarks. The horizontal bars rank the five numeric variables by their Pearson r with FinalMarks.

Observation: InternalMarks (0.600) has the strongest link, followed by StudyHours (0.542), AssignmentScore (0.510) and PreviousSemesterScore (0.472). Attendance (0.324) ranks last, so other academic behaviours are more closely tied to final results, although attendance is still positively related.

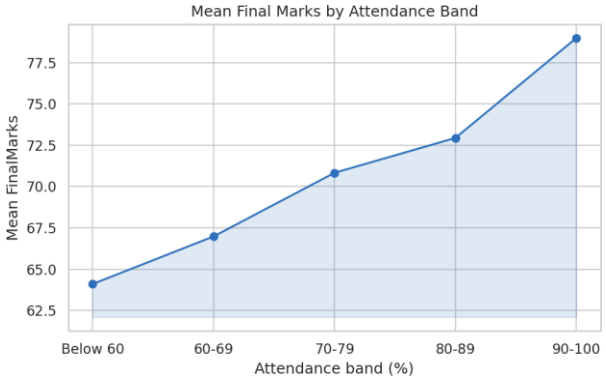


Figure 7. Mean FinalMarks by attendance band. The line plot shows average final marks for five attendance bands: below 60%, 60-69%, 70-79%, 80-89% and 90-100%.

Observation: Mean final marks increase at every step, from 64.1 (below 60%) to 67.0, 70.8, 72.9 and 79.0 (90-100%). The sharpest rise occurs between the 80-89% and 90-100% bands.

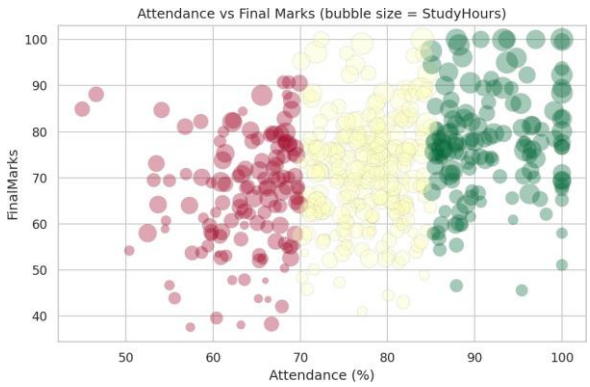


Figure 8. Attendance vs FinalMarks with bubble size showing StudyHours. Each bubble is one student; larger bubbles mean more study hours, and the colour changes from red (low attendance) to green (high attendance).

Observation: High-attendance students (green, right side) tend to have larger bubbles and higher marks, while low-attendance students (red, left side) tend to have smaller bubbles and lower marks. This supports the link between attendance and study habits (mean study hours 6.33, 6.76 and 7.46 for Low, Medium and High).

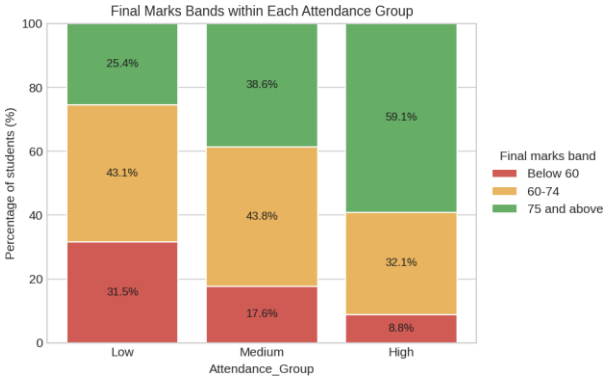


Figure 9. Final marks bands within each attendance group. Each stacked bar shows the percentage of students scoring below 60, 60-74 and 75 or above within a group.

Observation: The share of students scoring 75 or above rises from 25.4% in the Low group to 38.6% in Medium and 59.1% in High, while the share scoring below 60 falls from 31.5% to 17.6% and 8.8%. This confirms the positive pattern seen in the group means.

Summary of visual findings

- Every chart shows the same overall pattern: higher attendance goes with higher final marks (r = 0.324; group means 66.38, 71.06 and 76.96).
- The relationship is moderate, not strong: the points in Figure 2 are widely scattered and the group boxes in Figure 3 overlap.
- Attendance ranks last among the five variables in Figure 6, so InternalMarks, StudyHours and AssignmentScore are more closely tied to FinalMarks.
- Attendance is also linked with study hours (Figures 4 and 8), which may partly explain why regular attendees score higher. Because the data are observational, these charts show association, not cause.

6. Statistical Analysis (Basic)

a. Central tendency and spread

Table 1 summarises the mean, median and standard deviation for the numeric academic variables. Mean and median values are close for the variables shown in the EDA table. However, symmetry should be confirmed using the calculated skewness values; do not claim every variable is within ±0.2 unless the code output verifies that condition.

b. Correlation test results

Test / comparison	Result	Interpretation
Pearson r: Attendance, FinalMarks	r = 0.324; p = 1.2 × 10·¹³	Positive weak-to-moderate association
Pearson r: Attendance, InternalMarks	r = ·0.078; p = 0.083	Not significant at 5%
NumPy corrcoef / z-score check	r · 0.324 in both	Standardization preserves Pearson r
Coefficient of determination	r² = 0.324² · 0.105	About 10.5% shared linear variation

Table 5. Correlation results

c. One-way ANOVA across attendance groups

H·: mean FinalMarks are equal for Low, Medium and High attendance groups.
H·: at least one group mean differs.

ANOVA: FinalMarks	F(2, 497) = 26.43; p = 1.2 × 10·¹¹	Reject H· at 5%; at least one mean differs
ANOVA: InternalMarks	F = 1.10; p = 0.334	Fail to reject H·
Medium · Low mean FinalMarks	71.06 · 66.38 = 4.68	Medium mean is higher
High · Medium mean FinalMarks	76.96 · 71.06 = 5.90	High mean is higher
High · Low mean FinalMarks	76.96 · 66.38 = 10.58	Largest difference
Attendance ·75% vs <75%	74.02 vs 67.45; n=305 vs 195	About 6.6 marks higher in ·75% group

Table 6. ANOVA results and group mean comparison

Interpretation: Final marks increase from Low to Medium to High attendance groups. This agrees with the positive attendance·final-marks correlation. The InternalMarks test does not show a statistically significant group difference at the 5% level. ANOVA indicates whether any group mean differs; it does not identify which pairs differ. Check assumptions and use post-hoc tests for pairwise conclusions.
These supplied values should be confirmed by running the complete script on the exact CSV used in the report.

i. Conclusion & Future Scope

This project analyzed attendance and academic performance in 500 records. Descriptive analysis, visualizations, Pearson correlation, and one-way ANOVA showed a positive association between attendance and final marks and differences in mean final marks across attendance groups. Internal marks and study hours were also positively associated with final marks. Because the dataset is observational, causal conclusions cannot be drawn.

Future work: verify dataset provenance and attendance-group thresholds; analyze a larger representative dataset; check test assumptions; perform suitable post-hoc comparisons; and consider multivariable regression. Protect student privacy and avoid using one variable as the sole basis for decisions.

TEAM CONCLUSION TO COMPLETE: After rerunning the notebook, write a brief reflection on whether the objectives were met and one limitation to address next.

ii. References

1. Kaggle. Student Performance Dataset. [Insert exact verified URL and access date].
2. ITM University Gwalior. PBL Mini Project Report — Submission Format, Data Analytics Using Python (CSL0312), July–December 2026.
3. ITM University Gwalior. PBL Synopsis Approval Submission Format: Analysis of Student Attendance vs Academic Performance, dated 31 August 2026.
4. Python Software Foundation. Python Documentation. https://docs.python.org/3/
5. pandas development team. pandas documentation. https://pandas.pydata.org/docs/
6. SciPy community. SciPy documentation. https://docs.scipy.org/doc/scipy/

iii. AI Tool Usage Declaration

AI tool	Purpose	Extent / section applied
ChatGPT	Limited support with report structure and explanatory wording	Drafting report explanations; team verified code, calculations, charts, and conclusions
Other AI tools	NA	NA

iv. Project Links & Deliverables

Deliverable	Team entry / status
GitHub repository	[Insert repository URL]
Notebook (.ipynb)	[Insert exact filename]
Dataset	students_500_dataset_updated.csv
Report	Student_Attendance_vs_Academic_Performance_PBL_Report.docx / .pdf
README	[Confirm setup, dataset, notebook run instructions, outputs]
Repository checklist	[Confirm code, notebook, report, README, and dataset/licensing details]

v. Report Structure Checklist

No.	Required section	Status / action
1	Title page	✓
2	Introduction & problem statement	✓
3	Dataset description & source	✓
4	Data cleaning & preprocessing	✓
5	EDA	✓
6	Visualization & observations	✓
No.	**Required section**	**Status / action**
7	Statistical analysis	✓
8	Key insights	✓
9	Conclusion & future scope	✓

10	References	✓
11	AI declaration	✓
12	Project links	✓

vi. Declaration by Student(s)

We declare that this mini-project report is based on the dataset and analysis reviewed by our team. All data sources, references, and AI assistance have been acknowledged to the best of our knowledge. We are responsible for the accuracy of the submitted work and for explaining the code and results.

Team Leader: P. Chethan (BETN1A125194) Signature: _____

Team Member: G. Nandini (BETN1A125164) Signature: _____

Team Member: J. Durgeshwar (BETN1A125167) Signature: _____

Date: _____

vii. For Faculty Use Only

Evaluation component	Max marks	Marks awarded
Data Cleaning & Preprocessing	2	
Exploratory Data Analysis	2	
Visualization & Observations	2	
Statistical Analysis	2	
Insights, Interpretation & Conclusion	2	
Total	10	

Faculty remarks: _____

Faculty Mentor Signature: _____Date: _____